

Dr. Mohammad Manjur Alam (Manju)
Associate professor
Department of Computer Science and
Engineering
International Islamic University Chittagong.

CHAPTER 5



FUNDAMENTALS OF PROBABILITY



Concept related to probability

Experiment: Experiment is an act that can be repeated under given conditions.

Tossing of a coin or throwing of a dice or the drawing of a cards etc, are the example of experiment.

Outcomes: The results of an experiment are called outcomes.

Random experiment: Experiments are called random experiments if the outcomes depend on chance and cannot be predicted with certainty.

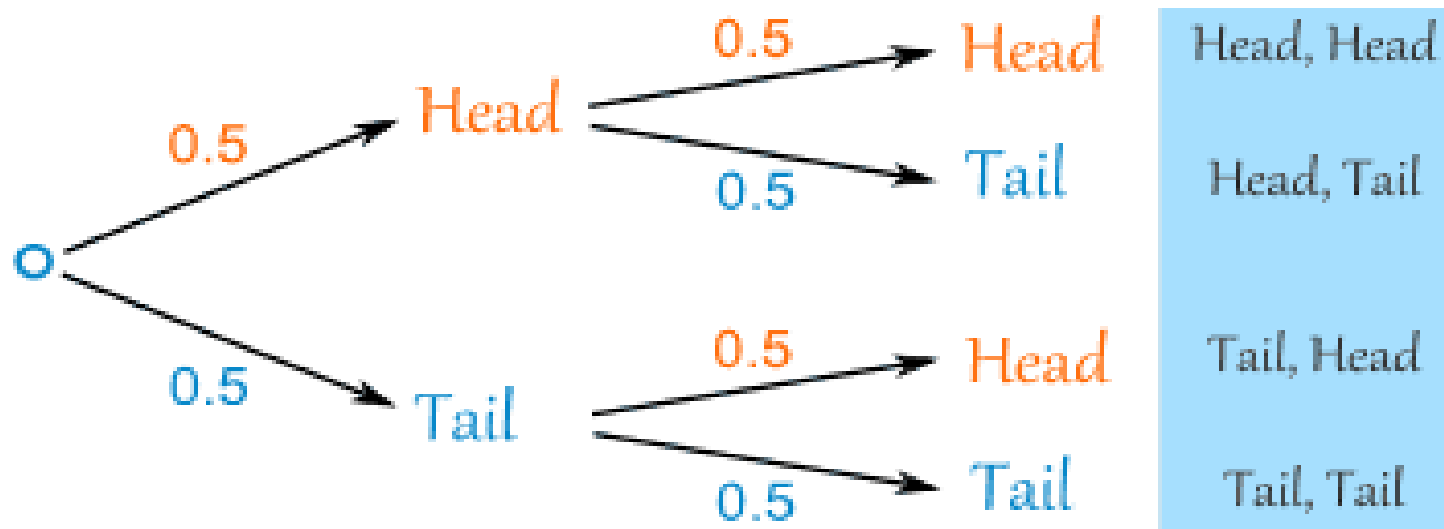
Example: Tossing of a fair coin, throwing of dice etc are the examples of random experiments.

Sample space: The collection or totality of all possible outcomes of a random experiment is called sample space. It is usually denoted by S or Ω .

where H and T denote the head and tail of the coin respectively.

If we toss a coin, the sample space is $S = \{H, T\}$

If we toss a coin two times, then the sample space is $S = \{HH, HT, TH, TT\}$



Sample point: Each element of a sample space is called sample point.

Event: Any subset of a sample space is called event.

Mutually exclusive events: Two events are said to be mutually exclusive if they have no common points. If A and B are two mutually exclusive events,

Example, $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$ then $A \cap B = \emptyset$

Non- mutually exclusive events: Two events are said to be non-mutually exclusive if they have common points. If A and B are two mutually exclusive events,

Example, $A = \{1, 2, 4\}$ and $B = \{2, 4, 6\}$ then $A \cap B = \{2, 4\}$

Conditional Probability: If A and B are two events in S. Then the conditional probability of A for given value of B, denoted by $P[A | B]$ is defined by

$$P[A | B] = P(AB) / P(B) ; P[B] > 0$$

Similarly, $P[B | A] = P(AB) / P(A) ; P[A] > 0$

Independent Event: Two events A and B are said to be independent if and only if one of the following conditions holds:

(I) $P[AB] = P[A]P[B]$

(II) $P[A | B] = P[A]$

(III) $P[B | A] = P[B]$

Dependent Event: Two events A and B are said to be dependent if and only if one of the following conditions holds:

(I) $P[AB] \neq P[A]P[B]$

(II) $P[A | B] \neq P[A]$

(III) $P[B | A] \neq P[B]$

Definition of probability:

There are four approaches of defining probability.

- Classical or mathematical or priori definition of probability.
- Empirical or statistical or posterior or frequency probability.
- Subjective probability.
- Axiomatic probability.

Classical or mathematical or priori definition of probability

If there are n mutually exclusive, equally likely and exhaustive outcomes of a random experiment and if m of these outcomes are favorable to an event A , then the probability of the event A , denoted by $P[A]$ is defined as



$$P(A) = \frac{N[A]}{N[S]} = \frac{m}{n} \quad ; 0 \leq P[A] \leq 1$$

This definition of probability is given by Laplace.

Axiomatic Probability

Suppose S is a sample space and A is an event of this sample space. Then the probability of the event A , denoted by $P[A]$ must satisfy the following four axioms:

(i) $P(A) \geq 0$;

(ii) $P(S) = 1$;

(iii) If A and B are mutually exclusive events, then $P[A \cup B] = P[A] + P[B]$;

(iv) Let $A_1, A_2, \dots, A_k$ be a sequence of K mutually exclusive events, then $P[A_1 \cup A_2 \cup \dots \cup A_k] = P[A_1] + P[A_2] + \dots + P[A_k]$

Laws of Probability : (i) Addition laws of probability

• (ii) Multiplication laws of probability.

Addition laws of probability are two types

(i) Mutually exclusive events

(ii) Non-Mutually exclusive events

Additive Law

If A and B are two events, then

(i) $P[A \cup B] = P[A] + P[B] - P[AB]$; Not Mutually Exclusive

(ii) $P[A \cup B] = P[A] + P[B]$; Mutually Exclusive

If A, B, C are three non **mutually exclusive** events then,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$$

If A, B, C are three **mutually exclusive** events then,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

Multiplication law of probability: If A and B are two events, then

$$P[AB] = P[A] P[B | A] = P[B] P[A | B]$$

Theorem: Show that the value of the probability lies between 0 to 1.

i.e

$$0 \leq P(A) \leq 1$$

Proof: Let A be the event in S (sample space)

Also let, $N(S) = n$

And $N(A) = m$

By the definition of probability, $P[A] = \frac{N(A)}{N(S)} = \frac{m}{n}$;

Clearly, $0 \leq m \leq n$

$$= \frac{0}{n} \leq \frac{m}{n} \leq \frac{n}{n}$$

$$= 0 \leq P(A) \leq 1$$

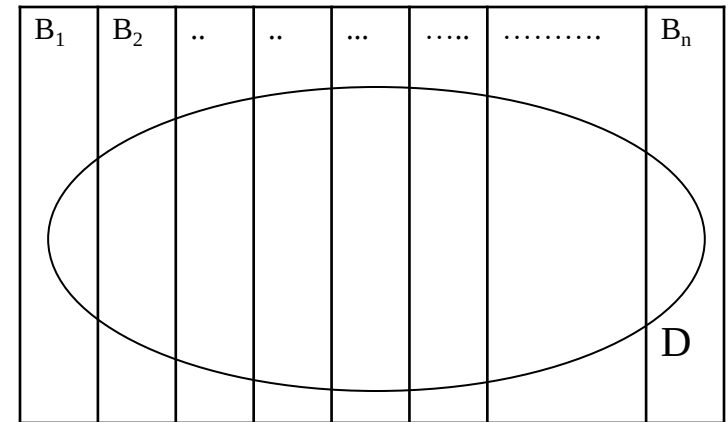
Theorem: If A and $\bar{A}$ are mutually exclusive then show that

$$P(A) + P(\bar{A}) = 1$$

Statement of Bayes theorem

Let $B_1, B_2, \dots, B_n$ be n mutually exclusive and exhaustive events in a random experiment and D be any event in S , then Bayes theorem state that

$$P[B_i|D] = \frac{P[B_i] \cdot P[D/B_i]}{\sum_{i=1}^n P[B_i] P[D/B_i]}; i = 1, 2, \dots, n$$



Mathematical problem of Bayes' theorem:

In a factory, machines A, B and C manufacture respectively 25%, 35% and 40% of the product. Of the total of their output 5, 4 and 2 percent are defective bolts respectively. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by machine (i) A and (ii) C.

Solution: Let

B_1 : An item produced by machine A

B_2 : An item produced by machine B

B_3 : An item produced by machine C

D: Defective item produced by the machines.

We have, $P[B_1] = 25\% = 0.25$, $P[B_2] = 35\% = 0.35$, $P[B_3] = 40\% = 0.40$,

And $P(D|B_1) = 5\% = 0.05$, $P(D|B_2) = 4\% = 0.04$, $P(D|B_3) = 2\% = 0.02$,

We have to find, $P(B_1|D)$ and $P(B_3|D)$

$$P[B_i|D] = \frac{P[B_i] \cdot P[D/B_i]}{\sum_{i=1}^n P[B_i] P[D/B_i]} ; i = 1, 2, \dots, n$$

$$P(B_1|D) = \frac{P(B_1)P(D/B_1)}{P(B_1)P(D/B_1) + P(B_2)P(D/B_2) + P(B_3)P(D/B_3)}$$

$$= \frac{(0.25)(0.05)}{(0.25)(0.05) + (0.35)(0.04) + (0.40)(0.02)}$$

$$= 0.3623$$

CHAPTER 6/7

RANDOM VARIABLE AND MATHEMATICAL EXPECTATION

A random variable is a real valued function whose values are determined with the outcomes of a random experiment. It is usually denoted by X, Y, Z etc

Let us consider the experiment of tossing two fair coins. The sample space of the experiment is

$$S = \{HH, HT, TH, TT\}$$

Let X denotes the number of heads. So,

Here, X can take the values 0, 1 and 2. Therefore X is a random variable.

Sample point	Number of Head (X)
HH	2
HT	1
TH	1
TT	0

Types of random variable.

Ans: There are two types of random variable:

1. Discrete random variable.
2. Continuous random variable.

Discrete random variable: A random variable is called discrete random variable if it can take only isolated values.

Example: Family members, mobile number etc.

Continuous random variable: A random variable is called continuous random variable if it can take any values between certain limits.

Example: Age, weight, height etc.

Probability function:

A function $f(x)$ of a discrete random variable X is called a probability function if it satisfies the following two conditions:

$$(i) f(x) \geq 0$$

$$(ii) \sum f(x) = 1$$

Probability density function:

A function $f(x)$ of a discrete random variable X is called a probability function if it satisfies the following two conditions:

$$(i) f(x) \geq 0$$

$$(ii) \int_{-\infty}^{\infty} f(x) dx = 1$$

Mathematical expectation of a random variable

If X is a discrete or continuous random variable with probability function or probability density function $f(x)$, then the mathematical expectation of X is usually denoted by $E[X]$ or μ and defined by

$$(i) \mu = E(X) = \sum xf(x) \\ = \int_{-\infty}^{\infty} xf(x)dx$$

Properties of mathematical expectation of a random variable:

- If b is a constant then $E[b] = b$
- If X is a random variable with expectation $E[X]$, then $E[aX+b] = aE[X] + b$, Where a and b constant.
- If X is a random variable with expectation $E[X]$, then $E[X-E(X)] = 0$
- If X and Y are random variables then $E[X+Y] = E[X] + E[Y]$
- If X and Y are random variables then $E[X-Y] = E[X] - E[Y]$

Properties of variance of a random variable:

- If b is a constant then $V[b] = 0$

- If X is a random variable with expectation $E[X]$, then

$$V(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2$$

- If X and Y are random variables then $V[X+Y] = V[X] + V[Y]$

- If X and Y are random variables then $V[X-Y] = V[X] + V[Y]$

- If X is a random variable, then $V[aX+b] = a^2 V[X]$, Where a and b constant.

Application problem: A discrete random variable X has the following probability function:

Values of X:x	0	1	2	3	4
f (x)	0.12	0.18	k	0.30	0.16

Find the value of k, Compute (ii) $P[X > 3]$; (iii) $P[1 < X < 4]$; and (iv) $P[X < 1]$.

Solution: (i) Since,

$$\sum f(x) = 1$$

$$\text{Or, } (0.12 + 0.18 + k + 0.30 + 0.16) = 1$$

$$\text{Or, } 0.76 + k = 1$$

$$\text{Or, } k = 1 - 0.76$$

$$\text{Or, } k = 0.24$$

$$(ii) P[X > 3] = P[X = 4] = 0.16$$

$$(iii) P[1 < X < 4] = P[X = 2] + P[X = 3] \\ = k + 0.30 \\ = 0.24 + 0.30 = 0.54$$

$$(iv) P[X < 1] = P[X = 0] \\ = 0.12$$

Assignment problem: A discrete random variable X has the following probability function:

Values of $X:x$	0	1	2	3	4
$f(x)$	0.10	0.30	0.20	0.25	0.15

Find the value of (i) $P[X = 1]$; (ii) $P[X > 3]$; (iii) $P[1 < X < 4]$; and (iv) $P[X < 1]$.

Problem: A continuous random variable X has the following probability density function:

$$f(x) = kx^2 ; 0 < x < 1$$

- i. Find the value of K
- ii. probability that X lies between 0.2 and 0.50
- iii. probability that X less than 0.30 and
- iv. probability that X greater than 0.75
- v. $E[x]$
- vi. $V[x]$
- vii. $SD[x]$

A continuous random variable X has the following probability density function:

$$f(x) = K(x+1); \quad 2 \leq x \leq 5$$

Compute (i) the value of 'K' ;

(ii) $P(X > 3)$;

(iii) $P(X = 4)$

(iv) $P(3 < X < 4)$

(v). $E[x]$

(vi). $V[x]$

(vii). $SD[x]$

CHAPTER 7/8

PROBABILITY DISTRIBUTION

□ Binomial Distribution

A discrete random variable X is said to have binomial distribution if its probability function is as follows:

$$f(x) = {}^n C_x p^x q^{n-x} ; x = 0, 1, 2, \dots, n$$

Where n and p are the parameters of the distribution and $p + q = 1$

□ Assumption of binomial distribution:

- The sample consists of n fixed observations.
- Each observation consist of two possible outcomes called success and failure.
- The probability of success, p , or a failure, $1 - p$, remain same from trial to trial.
- The outcome (success or failure) of any observation is independent of the outcome of any other observation.

❑ Poisson distribution

A discrete random variable X is said to have Poisson distribution if its probability function is as follows:

$$f(x) = \frac{e^{-m} m^x}{x!}; x = 0, 1, 2, \dots, \infty,$$

Where m is the parameter of the distribution and $e = 2.718$

❑ Some practical application of Poisson distribution:

- The number of Suicides reported in a particular day.
- The number of faulty blades in a packet of 100.
- The number of printing mistakes per page of a book.
- The number of letters lost in a mail per day.
- Number of deaths from a rare disease in a locality.

NORMAL DISTRIBUTION

A continuous random variable X is said to have normal distribution if its probability density function is as follows:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \begin{array}{l} ; \infty \leq x \leq \infty \\ ; \infty \leq \mu \leq \infty \\ ; \sigma^2 \geq 0 \end{array}$$

μ, σ^2 are the parameters of the distribution.

Standard normal variate

A variate is called standard normal variate if its mean 0 and variance 1 respectively. It is denoted by Z where, $Z = \frac{X - \mu}{\sigma}$

Properties of normal distribution:

- I. Normal distributions are symmetric around their mean.
- II. The mean, median, and mode of a normal distribution are equal.
- III. The area under the normal curve is equal to 1.0.
- IV. Normal distributions are denser in the center and less dense in the tails.
- V. Normal distributions are defined by two parameters, the mean (μ) and the standard deviation (σ).
- VI. 68% of the area of a normal distribution is within one standard deviation of the mean.
- VII. Approximately 95% of the area of a normal distribution is within two standard deviations of the mean.

I. Importance of normal distribution:

- II. In practice under certain condition most of the probability and sampling distributions can be approximated by normal distribution.
- III. According to central limit theorem, if mean and variance of a distribution exist, then the distribution converted to normal distribution.
- IV. Normal distribution is the basis of all the sampling distribution.
- V. Assumption of normality is the basis of all the test of significance in applied statistics.
- VI. Normal distributions find its application in industrial statistics such as quality control.

Application problem: A family has five children. The probability of a child being a boy or a girl is equal. Find the probability that the family has (i) no boy; (ii) at most 2 boys and (iii) at least 3 boys.

Solution:

Here, $n = 5$, $p = q = 0.50$ (since, The probability of a child being a boy or a girl is equal)

We Know, The probability function of a binomial distribution is

$$f(x) = {}^nC_x p^x q^{n-x} ; x = 0, 1, 2, \dots, n$$

Therefore,

$$f(x) = {}^5C_x (0.50)^x (0.50)^{5-x}$$

(i) The probability that the family has no boy i.e,

$$f(0) = {}^5C_0 (0.50)^0 (0.50)^{5-0} = 0.03125$$

(ii) The probability that at most 2 boys is

(iii) The probability that at least 3 boys is

Assignment problem: A fair coin is tossed 5 times. Find the probability of (i) exactly two heads, (ii) at least 3 heads, (iii) no heads and (iv) at most 2 heads.

CHAPTER 8

ESTIMATION AND TEST OF HYPOTHESIS

Hypothesis: A statement about the nature of a population.

Example: Students who eat breakfast will perform better on a stat exam than students who do not eat breakfast.

Test of Hypothesis: The statistical procedure which is used to verify any statement or assumption about population parameter on the basis of sample observations is known as test of significance.

Null Hypothesis: The hypothesis which we are going to test for possible rejection under the assumption that it is true. Null hypothesis is denoted by

$$H_0 ; H_0: \mu_1 = \mu_2$$

Alternative Hypothesis: Each of all possible hypothesis other than null hypothesis is called alternative hypothesis and is usually denoted by H_1 ; $H_0: \mu_1 \neq \mu_2$

Type-I error: The error of rejecting H_0 (accepting H_1) which is true is called error of first kind or Type-I error.

Type –II error: The error of accepting null hypothesis H_0 when it is false is called error of 2nd kind or H_1 . Type-II error is denoted by β

Level of significance: The probability of Type-I error denoted by α is called level of significance. Symbolically $\alpha = P(\text{rejecting } H_0 | H_0 \text{ is true})$. It is also known as size of a test

P –value:

The p value is the level at which the given value of the test statistic (such as t, F,) would be on the border line between the acceptance and rejection regions. The decision rules, which most researchers follow in stating their results, are as follows:

If $p < 0.01$, the results are regarded as highly significant

If $0.01 < p < 0.05$, the results are regarded as statistically significant

If $p > 0.10$, the results are considered not significant

Acceptance region:

The set of values of test statistic, which provides evidence to agree with the null hypothesis and lead to the acceptance of null hypothesis is called acceptance region.

Critical region:

A region of rejection is a set of possible values of the sample statistic, which provides evidence to contradict the null hypothesis and lead to a decision to reject the null hypothesis. It is denoted by

Power of a test:

The complement of the probability of Type-II error is called the power of a test. The probability of correct decision is called power of a test. Symbolically, $1 - \beta = P(\text{accepting } H_0 | H_0 \text{ is true}) = P(\text{rejecting } H_0 \text{ when } H_0 \text{ is false})$

One tailed test:

A one tailed test is a test which is concerned about possible deviation of the value of the parameter in only one direction from the specified value defined in the null hypothesis.

Example, $H_0: \mu = \mu_0$; $H_1: \mu > \mu_0$ or $\mu < \mu_0$

Two tailed test:

A two tailed test is a test which is concerned about the possible deviation of the parametric value in both the directions.

Example, $H_0: \mu = \mu_0$; $H_1: \mu \neq \mu_0$

Procedure of test of hypothesis

- i. **Set of hypothesis**
- ii. **Set up of suitable significance level**
- iii. **Determination of suitable test statistic**
- iv. **Determine the critical region**
- v. **Doing computation**
- vi. **Making decision**

COMMONLY USED TEST STATISTIC

1. The normal test (z-test):

- a. Testing the significance of population mean
 - i. Population variance known
 - ii. Population variance Unknown
- b. Testing the equality of two population means
- c. Testing the significance of single proportion
- d. Testing the equality of two Population proportions

2. The t test:

- a. To test the significance of population mean μ
- b. To test the significance in the difference between two population means
- c. To test the equality of two correlated means
- d. To test the significance of population correlation coefficient
- e. To test the significance of the regression coefficient
- f. To test the equality of two independent regression coefficients

3. Chi-square (χ^2) test

- a. To test the significance of a specified population variance
- b. To test the goodness of fit of a distribution
- c. To test the independence of attributes
- d. To test the equality of several correlation coefficients
- e. To test the homogeneity of several tests
- f. To test the homogeneity of several population variance

4. F-test:

- a. To test the equality of several population means
- b. To test the equality of two population variance
- c. To test the significance of regression
- d. To test the significance of multiple correlation coefficient



THANKS